

CSCI 5609 · ASSIGNMENT 01

Summer, on Screen

How a word in a title opens a window onto movie genres.

899	24	1946–2024	73 / 78
MOVIE TITLES	KNOWN GENRES	RELEASE YEARS	DRAMA TOP THREE
unique IMDb IDs	multi-label categories	one title has no year	most persistent genre

Two questions, two views

Q1 How do the annual top three genres change over time? Q2 Which genres most often appear together in a movie?

THE DATA The supplied summer_movies.csv contains 899 distinct titles whose names contain “summer.” A title may contribute to up to three genre counts. There are 24 known genre labels; 18 titles have no genre value. The distribution chart and co-occurrence matrix use every title with known genres. Annual rankings exclude the single title with an unknown release year.

THE METHOD For each year, genre counts are recalculated independently and the three highest are shown. Exact ties at the cutoff are ordered alphabetically. A year with no titles has no cells. The matrix counts titles that contain both row and column genres; its optional row-share view divides each pair count by the number of titles in that row genre.

Questions are answered for this title-filtered collection; they should not be read as claims about all films or the season in which a film was released. Co-occurrence describes association, not causation or a statistical correlation coefficient.

PROJECT LINKS Repository: github.com/jason-2333/A1 · GitHub Pages: jason-2333.github.io/A1

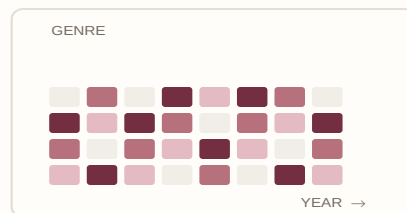
01 / QUESTION ONE

Which genres lead each year?

Three different encodings for annual genre counts and ranks. The selected design shows every genre over the full timeline.

A. Annual rank heatmap

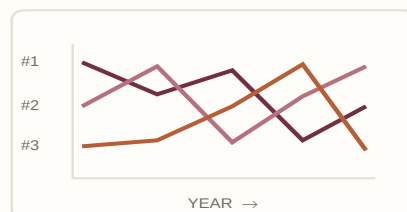
SELECTED



ENCODING Year → columns; genre → rows. Cell shade and rank number encode positions 1–3; blank cells mean the genre did not make that year's top three.

TRADE-OFF Aligned years and fixed genre rows make entries, exits, and persistence easy to compare across the full timeline.

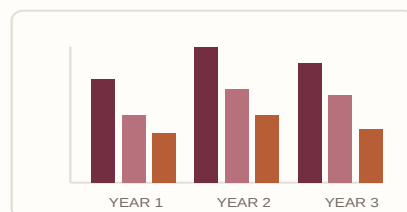
B. Rank bump chart



ENCODING Year → x; annual rank → y, with #1 at the top. Colored genre lines connect rank points and break when a genre leaves the top three.

TRADE-OFF Rank changes are immediate to see, but line crossings and repeated gaps become hard to follow over 79 years.

C. Grouped annual bars



ENCODING Year → groups; annual movie count → bar height; genre → color. Each year contains its leading three genre bars.

TRADE-OFF Movie counts are precise, but dozens of groups require more space and make long-term persistence hard to compare.

WHY THE HEATMAP It shows annual top-three membership across all years and all genres in one view, directly answering whether the leaders stay the same. Showing genre rows beyond the annual leaders makes turnover visible. Unlike the bump chart, it avoids tangled paths; unlike grouped bars, it keeps the full history compact.

FINDING The leading set changes: Drama, Fantasy, and History are the three listed in 1946; Drama, Comedy, and Music lead in 2024. Drama appears in the top three in 73 of 78 years with known genres, followed by Comedy in 61 and Romance in 33. No genre appears in every one of those years. Early years have sparse counts and ties, so those ranks are less stable.

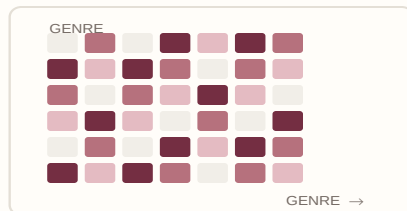
02 / QUESTION TWO

Which genres share a movie?

Three encodings for pairwise genre co-occurrence, with every known genre considered together.

A. All-genre co-occurrence matrix

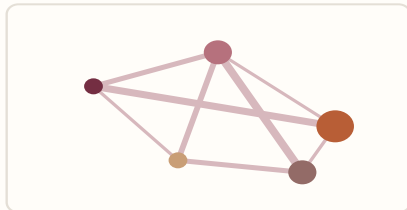
SELECTED



ENCODING Genre → both rows and columns. Each square represents a pair; shade and printed value encode the number of shared titles. A toggle shows row share.

TRADE-OFF All 24 genres remain visible and aligned for comparison. Numbers reveal exact counts; the row-share toggle helps compare proportions.

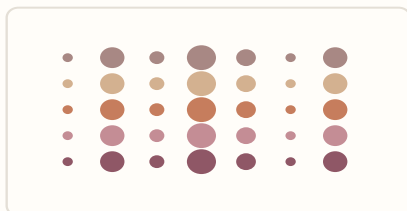
B. Weighted genre network



ENCODING Genre → node; node area → total titles in a genre. An edge means two genres share titles; edge width → pair count; color → genre.

TRADE-OFF Clusters can suggest groups, but edge crossings and movable nodes make pair counts difficult to compare precisely.

C. Bubble co-occurrence matrix



ENCODING Genre → both axes. Circle size → pair count; color → row genre. A size key maps bubble area to movie count.

TRADE-OFF The grid keeps all pairs together, but small circles are hard to compare and area is less precise than a labeled value.

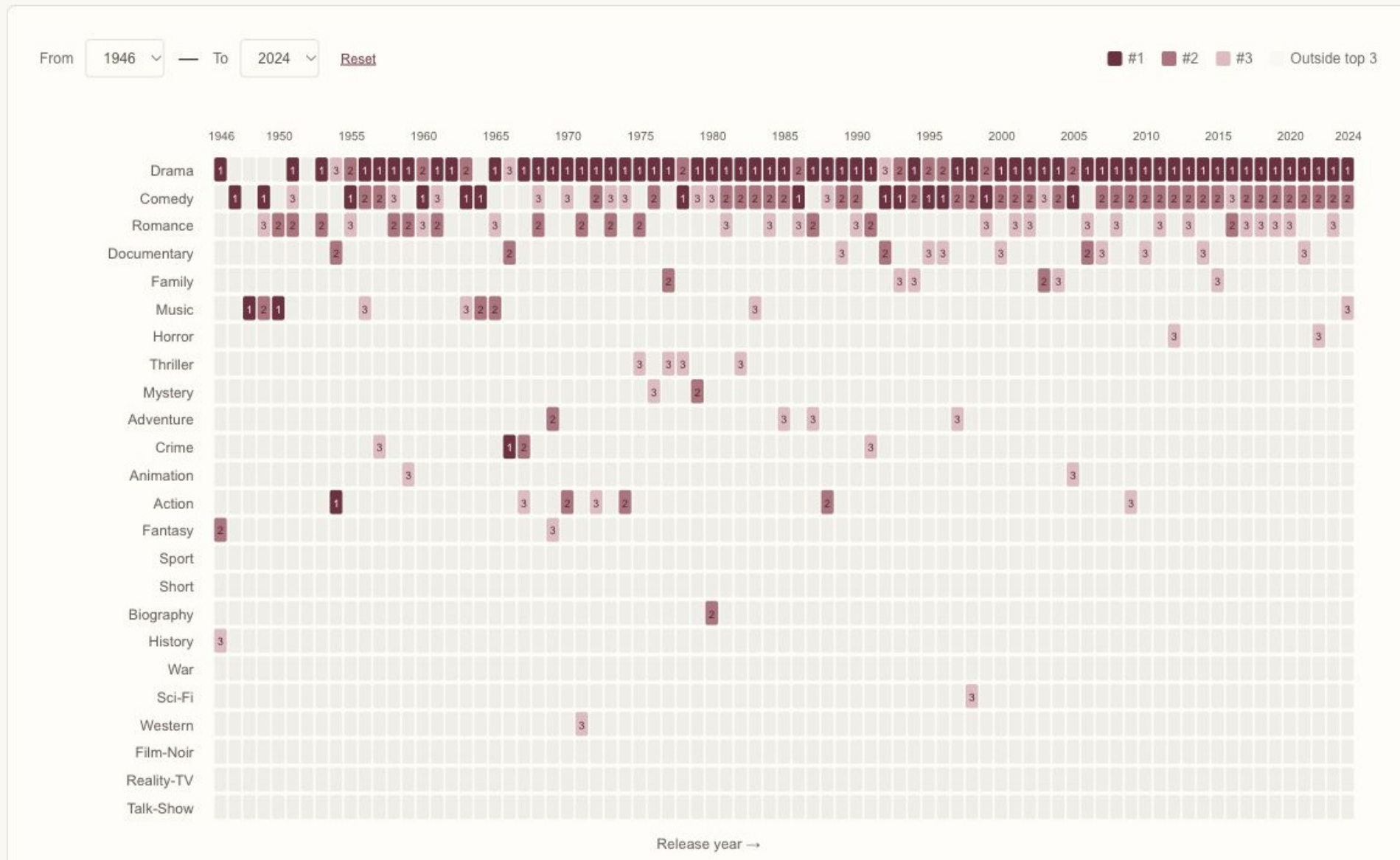
WHY THE MATRIX It places every genre in a single view, as the assignment asks, and aligned rows and columns support direct comparisons. Printed counts preserve the difference between 108 movies and 73; color makes strong pairs visible at a glance. A network layout depends on changing node positions, while small bubbles obscure low-count pairs. The row-share toggle gives a separate, clearly labeled proportional comparison.

FINDING Comedy co-occurs most with Drama: 108 titles, or 40% of the 270 known-genre Comedy titles. Romance follows with 73 (27%), then Family with 33 (12%). Counts give common genres an advantage; row shares help compare the proportion within each genre. Percentages for rare genres may be based on only a few titles.

03 / IMPLEMENTED VIEW

Annual top-three genres, 1946–2024

Rank is recalculated within each release year. Select a year range or inspect any cell for the genre count.

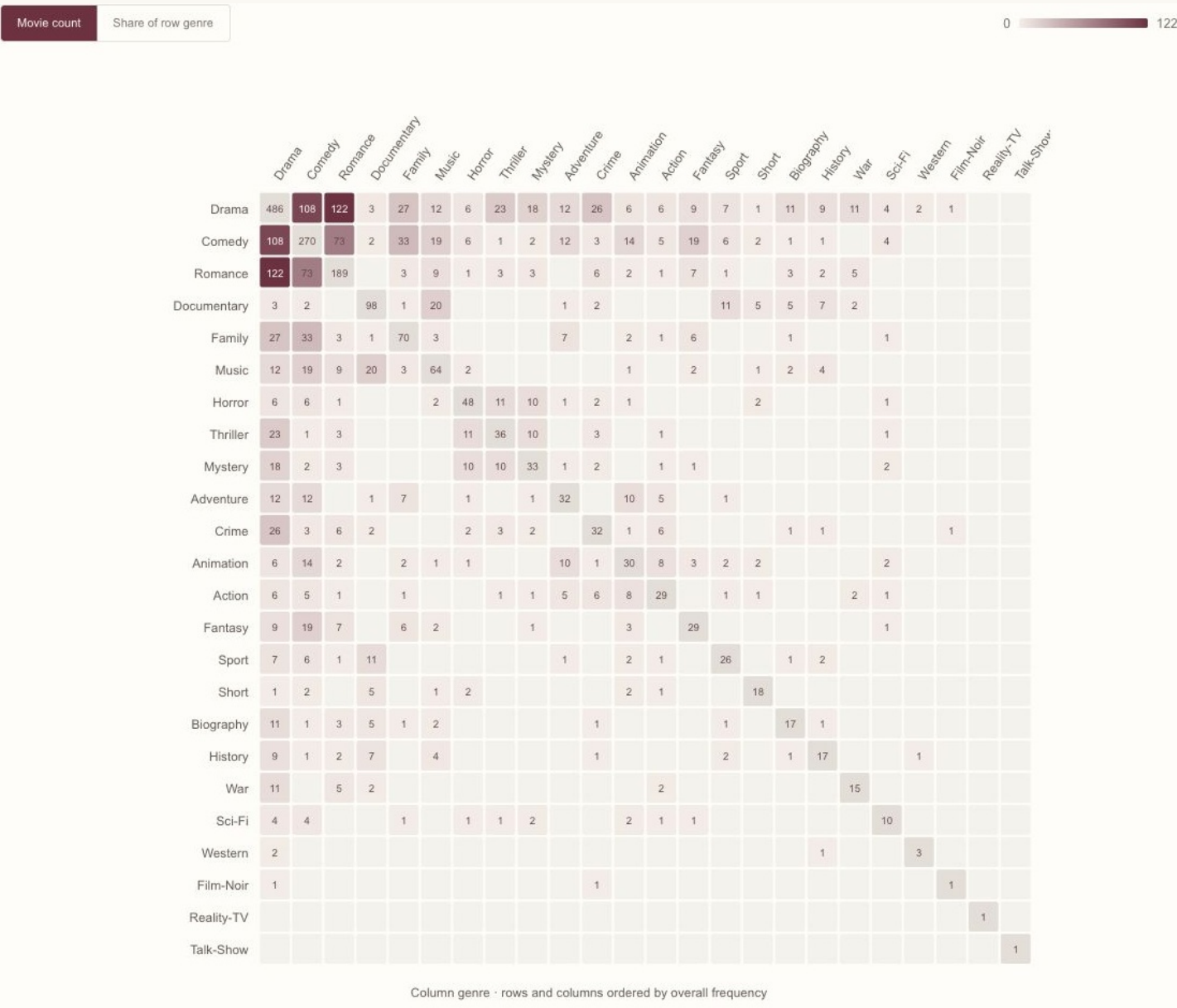


HOW TO READ Rows are genres, columns are release years, and darker cells indicate a higher annual rank. The number inside the cell gives the rank. Blank cells fall outside the top three. The chart orders tied counts alphabetically; 1952 has no titles, and the one title with unknown year is excluded.

04 / IMPLEMENTED VIEW

Genre co-occurrence across the collection

All 24 known genres in one matrix. The initial view shows shared-title counts; the page also provides a row-share view.



HOW TO READ Each off-diagonal cell counts titles with both genres. The diagonal shows each genre's total. Use the count/share control on the webpage to switch measures; row share divides a pair count by its row genre's total. Comedy + Drama = 108 titles (40% of Comedy); Comedy + Romance = 73 (27%); Comedy + Family = 33 (12%).

SOURCES Dataset: Summer Movies CSV supplied in the course assignment instructions, derived from IMDb non-commercial data. Calculations and screenshots were checked against the supplied CSV.